HW 0317

3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = \underset{(7.383)}{6.855} + \underset{(0.112)}{3.880} INCOME \quad \text{(se)}$$

a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.

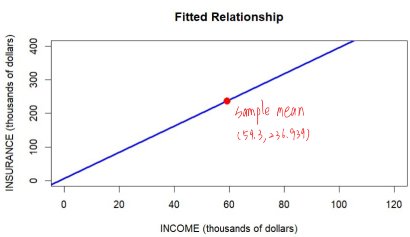
b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is −0.746.

d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

a.



estimated slope : 3.88, intercept : 6.855

sample mean $= 6.855 + 3.88 \times 59.3 = 236.939$

b. point estimate : the average amount of insurance will increase $3.88 \times 1 = 3.88$ with each additional 1000 of household income.

95% interval estimate :

$b_1 = 3.88$, two-tailed $\alpha = 0.05$, $df = 20-2 = 18 \Rightarrow t_c = 2.101$

CI: $b_1 \pm t_c se(b_1) \Rightarrow [3.88 - 2.101 \times 0.112, 3.88 + 2.101 \times 0.112]$

$= [3.645, 4.115]$

$\Rightarrow$ In repeated sampling, about 95% amount of insurance held changes with each additional 1000 of household income will fall between increase 3.645 and 4.115.

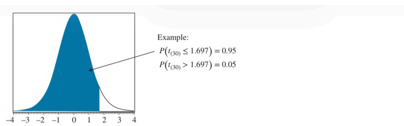


TABLE D.2 Percentiles of the *t*-distribution

df	$t_{(0.90, df)}$	$t_{(0.95, df)}$	$t_{(0.975, df)}$	$t_{(0.99, df)}$	$t_{(0.995, df)}$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	b. 2.101	e. 2.552	c. 2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797

c. $\widehat{insurance} = 6.855 + 3.88 \times 100 = 394.855$

$\sqrt{var(b_1 + 100 b_2)} = \sqrt{Var(b_1) + 100^2 Var(b_2) + 2 \times 100 \, cov(b_1, b_2)}$

$= \sqrt{7.383^2 + 10000 \times 0.112^2 + 200 \times (-0.746)}$

$= \sqrt{30.748689} = 5.5452$

99% interval estimate :

$\widehat{insurance} = 394.855$, two-tailed $\alpha = 0.01$, $df = 20-2 = 18 \Rightarrow t_c = \pm 2.878$

CI: $b_1 \pm t_c se(b_1) \Rightarrow [394.855 - 2.878 \times 5.5452, 394.855 + 2.878 \times 5.5452]$

$= [378.8959, 410.8141]$

$\Rightarrow$ In repeated sampling, about 99% interval constructed this way will contain the true value of the parameter.

d. Null hypothesis $H_0: \beta_2 = 5$ (For every \$1000 increase in income, the average amount of life insurance held increase by \$5000)

Alternative hypothesis $H_1: \beta_2 \neq 5$ (The statement isn't true)

Test statistic: $t = \frac{b_2 - \beta_2}{se(b_2)} = \frac{3.88 - 5}{0.112} = -10$.

Significant level: 5%, critical value for a two-tailed test with $df = 20 - 2 = 18$ is ± 2.101.

⇒ Reject H_0 when t-statistic falls outside the region of $-2.101 \sim 2.101$.

Here, $t = -10$ falls outside the rejection region

⇒ H_0 is rejected. The data do not support the claim.

e. Null hypothesis $H_0: \beta_2 = 1$ (The average amount of life insurance held increase by the same amout as income increases.)

Alternative hypothesis $H_1: \beta_2 \neq 1$ (The statement isn't true)

Test statistic: $t = \frac{b_2 - \beta_2}{se(b_2)} = \frac{3.88 - 1}{0.112} = 25.7143$

Significant level: 1%, critical value for a two-tailed test with $df = 20 - 2 = 18$ is ± 2.552

⇒ Reject H_0 when t-statistic falls outside the region of $-2.552 \sim 2.552$

Here, $t = 25.7143$ falls outside the rejection region

⇒ H_0 is rejected. The slope is larger than 1

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SOFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

c. Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

a.

```
> #estimated model
> model = lm(price ~ I(sqft^2), data = collegetown)
> summary(model)

Call:
lm(formula = price ~ I(sqft^2), data = collegetown)

Residuals:
    Min      1Q  Median      3Q     Max 
-383.67  -48.39   -7.50   38.75  469.70 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 93.565854   6.072226   15.41   <2e-16 ***
I(sqft^2)    0.184519   0.005256   35.11   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.08 on 498 degrees of freedom
Multiple R-squared:  0.7122,	Adjusted R-squared:  0.7117 
F-statistic:  1233 on 1 and 498 DF,  p-value: < 2.2e-16
```

⇒ estimated model: $\widehat{PRICE} = 93.5659 + 0.1845\, SQFT^2$

```
> # marginal effect and std
> alpha2 = coef(model)[[2]]
> marginal_effect = 2*alpha2*20
> sd = 40*(0.005256)
```

$= \frac{dPRICE}{dSQFT} = 2\hat{\alpha}_2 SQFT$

$= \sqrt{\widehat{var}(40\hat{\alpha}_2)} = 40se(\alpha_2)$

$H_0: 2\alpha_2 SQFT \le 13.$

$H_1: 2\alpha_2 SQFT > 13.$

right-tail test.

```
> # test
> t = (marginal_effect-13)/sd
> df = dim(collegetown)[1]-2
> p = 1 - pt(q = t, df = df, lower.tail = T)
> print(paste("t-value =", t))
[1] "t-value = -26.7277391681723"
> print(paste("df =", df))
[1] "df = 498"
> print(paste("p-value =", p))
[1] "p-value = 1"
```

→ test statistic.

⇒ $t_{N-2=498}$

critical value. $t_{0.95,498} = 1.6479.$

⇒ rejection region: reject H_0 when test statistic > 1.6479

$\because$ t-value $= -26.728 < 1.6479$ ⇒ H_0 isn't rejected.

⇒ We can't conclude that the marginal effect is greater than \$13000.

b.

```
> # test the marginal effect
> sqft_2 <- 40
> marginal_effect_2 <- 2*qrmodel_A2*sqft_2
> se <- (qrmodel_A2_se*2*sqft_2)
> t_statistic_2 <- (marginal_effect_2 - 13)/se
> critical_2 <- qt(0.95, df)
> pvalue_2 <- 1-pt(t_statistic_2,df)
> print(paste("marginal effect =", marginal_effect_2))
[1] "marginal effect = 14.7615202345669"
> print(paste("se =", se))
[1] "se = 0.420464072040742"
> print(paste("t-statistic =", t_statistic_2))
[1] "t-statistic = 4.18946671475947"
> print(paste("critical value =", critical_2))
[1] "critical value = 1.64791913885501"
> print(paste("p-value =", pvalue_2))
[1] "p-value = 1.65394980079503e-05"
```

$= 2 \times 40 \times \hat{\alpha}_2$

$H_0: 2\alpha_2 SQFT \le 13.$

$H_1: 2\alpha_2 SQFT > 13.$

right-tail test.

critical value. $t_{0.95,498} = 1.6479.$

⇒ rejection region: reject H_0 when test statistic > 1.6479

$\because$ t-value $= 4.189 > 1.6479$ ⇒ H_0 is rejected.

⇒ We can conclude that the marginal effect is greater than \$13000.

c.

```
> sqft <- 20
> alpha <- 0.05
> tc <- qt(1-alpha/2,df)
> alpha1 <- model$coef[1]
> alpha2 <- model$coef[2]
> vara1 <- vcov(model)[1, 1]
> vara2 <- vcov(model)[2, 2]
> cova1a2 <- vcov(model)[1, 2]
> varL = vara1 + (sqft^2)^2*vara2 + 2*sqft^2*cova1a2 # var(L)
> seL <- sqrt(varL)
> exp_price <- alpha1 + alpha2*sqft^2
> print(paste("expected price =", exp_price))
[1] "expected price = 167.373455273212"
>
> # Interval estimate
> lowb <- exp_price-tc*seL
> upb <- exp_price+tc*seL
> interval<-c(lowb,upb)
> interval
(Intercept) (Intercept)
   158.0481    176.6988
```

$se = \sqrt{var(\hat{\alpha}_1)+400\hat{\alpha}_2}$

$= \sqrt{var(\hat{\alpha}_1)+400^2\hat{\alpha}_2+2\times 400\,cov(\hat{\alpha}_1,\hat{\alpha}_2)}$

$= 4.7464$

$= 93.5659 + 0.1845 \times 20^2$

$= 167.3735$

critical value: $t_{0.975,498} = 1.9647$

$\Rightarrow$ 95% interval estimate: $167.3735 \pm 1.9647 \times 4.7464 \Rightarrow [158.0481, 176.6988]$

$\Rightarrow$ In repeated sampling, about 95% amount of expected prices will fall between 158.0481 and 176.6988.

d.

```
> price<- collegetown$price
> str(subset(collegetown,sqft == 20)$price)
 num [1:3] 138 169 183
```

$\Rightarrow$ The houses sell for 138000, 169000, 183000 respectively.

```
> summary(subset(collegetown,sqft == 20)$price)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  138.0   153.5   169.0   163.3   176.0   183.0
```

It is in the estimated interval [158.0481, 176.6988]

Made with Goodnotes

3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable $SAL1$ = unit sales of brand no. 1 canned tuna, and $APR1$ = price per can of brand no. 1 tuna (in dollars).

a. Calculate the summary statistics for $SAL1$ and $APR1$. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus $WEEK$. How much variation in sales and price is there from week to week?

b. Plot the variable $SAL1$ (y-axis) against $APR1$ (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?

c. Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

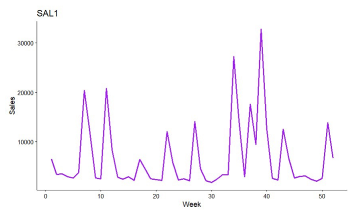
d. Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

e. Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

f. Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p-value for the test. What is your conclusion?

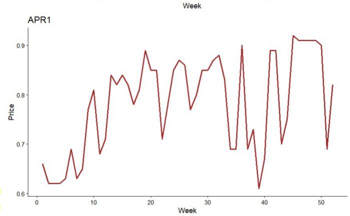
a.

```
> # SAL1
> mean_SAL1 <- mean(tuna$sal1)
> min_SAL1  <- min(tuna$sal1)
> max_SAL1  <- max(tuna$sal1)
> sd_SAL1   <- sd(tuna$sal1)
>
> # APR1
> mean_APR1 <- mean(tuna$apr1)
> min_APR1  <- min(tuna$apr1)
> max_APR1  <- max(tuna$apr1)
> sd_APR1   <- sd(tuna$apr1)
>
> # summary statistics
> summary_table <- data.frame(
+     Variable = c("SAL1", "APR1"),
+     Mean = c(mean_SAL1, mean_APR1),
+     Min = c(min_SAL1, min_APR1),
+     Max = c(max_SAL1, max_APR1),
+     SD = c(sd_SAL1, sd_APR1)
+ )
>
> print(summary_table)
  Variable      Mean     Min      Max           SD
1     SAL1 6718.7115 1762.00 32820.00 6.915083e+03
2     APR1    0.7825    0.61     0.92 9.803711e-02
```



Fluctuate significantly.

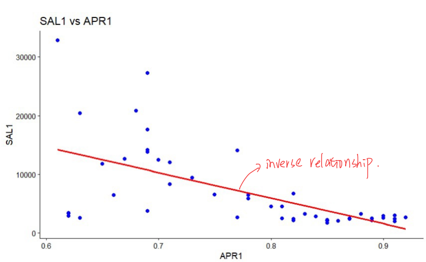
⇒ SAL1 versus WEEK has more variation, and its standard deviation is larger. (6915 units)



Fairly stable.

⇒ APR1 versus WEEK has less variation, and its standard deviation is smaller. (0.098 dollars)

b.



inverse relationship.

It's what we expected.

The law of demand: When the price is higher, the demand will be lower and the sales will be lower; and vice versa.

c.

```
> SAL1 <- tuna$sal1
> APR1 <- tuna$apr1
> PRICE1 <- 100*APR1
> linear_model <- lm(SAL1~PRICE1)
> summary(linear_model)

Call:
lm(formula = SAL1 ~ PRICE1)

Residuals:
     Min       1Q   Median       3Q      Max
-10869.5  -1588.3   -499.3   1626.2  18607.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
PRICE1       -434.45      78.58  -5.528 1.17e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,    Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06

> sum_linear <- summary(linear_model)
> b1 <- coef(sum_linear)[1]
> b2 <- coef(sum_linear)[2]
> se_b2 <-coef(sum_linear)[4]
> df <- sum_linear$df[2]
> tc <- qt(0.975,df)
> lowb <- b2-tc*se_b2
> upb <- b2+tc*se_b2
> print(paste("lower bound=",lowb))
[1] "lower bound= -592.289744663076"
> print(paste("upper bound=",upb))
[1] "upper bound= -276.604935808183"
```

$SAL1 = 40714.22 - 434.45 \times PRICE1$

point estimator for b_2 is -434.45

95% interval estimate: $[-592.2897, -276.6049]$

d.

```
> model <- lm(SAL1 ~ APR1, data = tuna)
> #price=0.70
> new_data <- data.frame(APR1 = 0.70)
> # 90% CI
> predict(model, newdata = new_data,
+         interval = "confidence", level = 0.90)
       fit      lwr      upr
1 10302.9 8624.934 11980.87
```

expected: 10302.9

90% CI: [8624.934, 11980.87]

e.

$SAL1 = 40714.22 - 434.45 \times PRICE1$, $\beta_2 = -434.45$

```
> # mean
> Pbar <- mean(APR1)  = 78.25
> Qbar <- mean(SAL1)  = 6718.71
> b1 <- coef(model)["APR1"]
>
> #elasticity at means
> elasticity <- b1 * (Pbar / Qbar)
> se_b1 <- summary(model)$coefficients["APR1", "Std. Error"]
> se_elasticity <- se_b1 * (Pbar / Qbar)
>
> #critical point
> n <- length(SAL1)
> tc_2 <- qt(0.975, df = n-2)
>
> #CI
> lowb_2 <- elasticity - tc_2 * se_elasticity
> upb_2 <- elasticity + tc_2 * se_elasticity
> print(paste("elatiscity=",elasticity))
[1] "elatiscity= -5.05982496477627"
> print(paste("lower bound=",lowb_2))
[1] "lower bound= -6.89814888681741"
> print(paste("upper bound=",upb_2))
[1] "upper bound= -3.22150104273513"
```

$= \frac{\Delta y / y}{\Delta x / x} = \beta_2 \frac{PRICE1}{SAL1} = \varepsilon$

$se(\varepsilon) = se(\beta_2) \times \frac{PRICE1}{SAL1} = 0.9152$

$\varepsilon \pm t_{\frac{\alpha}{2}, n-2} \times se(\varepsilon)$

elatiscity = −5.0598 → sales are highly sensitive to price changes.

This make economic sense, because canned tuna has many substitutes,

so the elatiscity is high.

f.

```
> #mean
> Pbar <- mean(APR1)
> Qbar <- mean(SAL1)
> #estimate
> b1 <- coef(model)["APR1"]
> se_b1 <- summary(model)$coefficients["APR1", "Std. Error"]
> # H0
> b1_h0 <- -3 * (Qbar / Pbar)
>
> #test
> t_statistic <- (b1 - beta1_null) / se_b1
> n <- length(SAL1)
> df <- n-2
> p_value <- 2 * (1 - pt(abs(t_stat), df))
> tc_3 <- qt(0.95, df)
>
> print(paste("t-statistic=", t_statistic))
[1] "t-statistic= -2.25057192191693"
> print(paste("critical value=",tc_3))
[1] "critical value= 1.67590502516311"
> print(paste("p-value=", p_value))
[1] "p-value= 0.0288420363284798"
```

< 0.1

H_0: elatiscity $= -3$.   two-tailed test.

H_0: elatiscity $\neq -3$.

test statistic: $t = \frac{elatiscity - (-3)}{se(elaticity)} \sim t_{N-2=50.} = -2.2506$

$\alpha = 0.1$, critical value: $|t_{0.95,50}| \Rightarrow |1.6759|$

⇒ rejection region: reject H_0 when test statistic doesn't fall in $[-1.6759, 1.6759]$

∵ t-value $= -2.2506 < -1.6479 \Rightarrow H_0$ is rejected.

⇒ We can conclude that the elasticity may not be −3.